SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 2.2: Area of Polygons — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.2: Area of Polygons
Driving Question	DRIVING QUESTION: How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon & Area of a Triangle (½ × base × height)**	Learners examined a sketch of an irregularly shaped farm plot (shamba) in Nakuru and noted that its boundary cannot be measured with a single rectangle or square. They observed that the plot could be roughly divided into triangular sections, prompting the question of how to find the area of each triangle accurately. Physical manipulation of cut-out triangles (formed by halving rectangles) provided concrete evidence that every triangle occupies exactly half the area of its enclosing rectangle.	The area of any triangle is calculated using A = ½ × base × height, where the HEIGHT must be the perpendicular (vertical) distance from the base to the opposite vertex — NOT the length of a slanted side. Learners re-established this foundational formula as the non-negotiable entry point for all polygon area work in this sub-strand. Pedagogical note: Expect common prior misconceptions — many learners will initially use a slant side as the 'height'; use targeted cold-calling to surface and correct this before proceeding.	The perpendicular height is essential because it represents the true vertical span of the triangle. When a triangle is 'leaned over' (obtuse or scalene), the slant side is longer than the true height, so using it would overestimate the area. Teachers should reinforce this by having learners measure both the slant side and the true perpendicular height on the same triangle and compare the two area calculations. Connect back to the driving question: a Nakuru land surveyor must always identify the correct height before applying any triangle area formula.
Lesson 2 **Area of a Triangle Using the Sine Formula (½ab sin θ)**	Learners observed a scenario in which a surveyor in Nakuru can measure two boundary fence lines (sides a and b) meeting at a corner peg, and can read the angle between them with a theodolite, but cannot physically measure the perpendicular height because crops or terrain block access. A geometric diagram showed how the perpendicular height h can be expressed as h = b sin θ, linking the familiar ½ × base × height formula to the	When the perpendicular height is not directly measurable but two sides and the included angle are known, the area of a triangle can be calculated using A = ½ab sin θ (equivalently written as ½ab sinC). This formula is derived directly from A = ½ × base × height by substituting h = b sinC. Learners must identify the INCLUDED angle — the angle sandwiched between the two known sides — to apply this formula correctly. Pedagogical note: Insist learners	The sine of the included angle acts as a scaling factor that converts the slant side into the equivalent perpendicular height. Because sin 90° = 1, when the angle is a right angle the formula reduces to the familiar ½ × base × height, confirming consistency. Teachers should demonstrate this limiting case explicitly. Connect to the driving question: a surveyor who can measure two fence lengths and the angle at a corner peg now has everything needed to

	new sine-based formula.	label which angle is 'included' before substituting values; mislabelling is the most frequent error at this stage.	find the triangle's area — no perpendicular tape measure required.
Lesson 3 **Applying the Sine Formula for Area of a Triangle**	Learners worked through a series of graded practice problems set in Kenyan contexts — a triangular maize plot near Eldoret, a triangular fishing ground on Lake Victoria, and a triangular tea estate parcel in Kericho. In each case, two sides and the included angle were provided. Learners observed how changing the included angle (while keeping the two sides fixed) dramatically changes the computed area, reinforcing that the angle is as important as the side lengths.	Learners consolidated fluency with $A = \frac{1}{2}ab \sin C$ through varied application. Key procedural steps are: (1) identify the two known sides; (2) confirm the angle given is the INCLUDED angle between those two sides; (3) substitute into the formula; (4) evaluate sin of the angle using a calculator or trigonometric tables; (5) round to appropriate significant figures and attach correct units (m^2, ha, acres). Learners also learned that if the given angle is obtuse, $\sin \theta$ is still positive, so the formula works without modification. Pedagogical note: Provide a structured worked-example template on the board and require learners to mirror that layout in their exercise books.	Consistent procedural layout prevents sign errors and unit omissions, which are the two most penalised mistakes in KCSE examinations. The reason the formula works for obtuse angles is that $\sin \theta = \sin(180° - \theta)$, so both an acute angle and its obtuse supplement yield the same sine value — and therefore the same area — confirming that a triangle and its 'mirror' have equal areas. Teachers should briefly address this to deepen understanding. Connect to the driving question: with this skill, a surveyor can handle any triangular sub-plot of the Nakuru shamba where a direct height measurement is impossible.
Lesson 4 **Area of a Triangle Using Heron's Formula**	Learners were presented with a triangular plot for which only the three boundary lengths (a, b, c) are known — no angles, no perpendicular height. They observed that neither $A = \frac{1}{2}bh$ nor $A = \frac{1}{2}ab \sin C$ can be applied directly without additional information. A historical note on Heron of Alexandria was connected to the fact that Kenyan surveyors today routinely use GPS to measure boundary distances without angles, making Heron's formula highly practical.	Heron's formula states: $A = \sqrt{[s(s-a)(s-b)(s-c)]}$, where $s = (a + b + c)/2$ is the semi-perimeter. The two-step procedure is: (1) ALWAYS calculate s first; (2) substitute s, a, b, c into the square-root expression. Learners must understand that s is not the perimeter — it is exactly half the perimeter. Skipping or mis-calculating s is the single most common error. Heron's formula produces the exact area using only the three side lengths, with no angle measurement required. Pedagogical note: Require learners to write out the s calculation as a separate, labelled line before proceeding; do not allow mental calculation of s.	Heron's formula is algebraically equivalent to $A = \frac{1}{2}bh$ but expressed entirely in terms of side lengths. It is derived by combining the cosine rule (to find an angle) with the sine area formula, then simplifying — a derivation teachers may share as enrichment for high-attaining learners. The formula is especially powerful for surveyors who use measuring chains or GPS to record distances, since no angle instrument is needed. Connect to the driving question: a Nakuru surveyor who has walked and measured all three sides of a triangular sub-plot can now compute its area with confidence using Heron's formula.
Lesson 5 **Polygons: Classification, Properties, and Decomposition into Triangles**	Learners were given a pentagon-shaped shamba outline (five boundary pegs) and asked to count how many non-overlapping triangles can fill it exactly, using only straight lines connecting existing vertices (diagonals from one vertex). They observed that drawing diagonals from a single chosen vertex always produces exactly $(n - 2)$ triangles, regardless of which vertex is chosen — for a pentagon, that is 3	Any polygon with n sides can be decomposed into exactly $(n - 2)$ non-overlapping triangles by drawing all diagonals from one vertex. This is the universal strategy for finding the area of any polygon: decompose → find each triangle's area using whichever formula is appropriate → sum all triangle areas. Key vocabulary consolidated: regular vs. irregular polygon, convex vs. concave, vertex, diagonal,	The $(n - 2)$ triangle rule is a direct consequence of the fact that each new diagonal from the chosen vertex 'uses up' one side of the remaining unfilled region, reducing the problem by one triangle at a time. Understanding this structure means a learner never needs to memorise separate formulas for hexagons, heptagons, and so on — decomposition into triangles is always available as a fallback strategy. Connect to

	triangles. Learners also observed that this works for convex polygons and were briefly shown that concave polygons require care.	interior angle. Learners also reviewed interior angle sums: $(n - 2) \times 180°$. Pedagogical note: Have learners physically draw and shade each triangle in a different colour on their polygon sketches to make the decomposition visible before any calculation begins.	the driving question: the Nakuru surveyor's irregularly shaped shamba, regardless of how many boundary pegs it has, can always be triangulated and each triangle solved independently.
Lesson 6 **Area of Regular Polygons: Central Triangle Decomposition**	Learners examined a regular hexagonal plot (such as a model of a beehive farming enclosure) and observed that when the centre point is connected to every vertex, the polygon splits into n congruent isosceles triangles, each with a central angle of 360°/n and two sides equal to the circumradius. They measured the apothem (perpendicular distance from centre to a side) and confirmed it is the height of each central triangle, enabling direct use of $A = ½ \times \text{base} \times \text{height}$ for each slice.	For a regular polygon with n sides of length s, the total area is given by $A = (n \times s^2) / (4 \times \tan(180°/n))$. This formula is derived by computing the area of one central triangle ($A_{triangle} = ½ \times s \times \text{apothem}$, where $\text{apothem} = s / (2 \tan(180°/n))$) and multiplying by n. Learners must be able to both apply the consolidated formula and reconstruct it from the central-triangle decomposition. Knowing n and s is sufficient to calculate the area of any regular polygon. Pedagogical note: Insist learners verify their formula result by also computing $n \times (½ \times s \times \text{apothem})$ to build formula-derivation confidence.	The formula works because all n central triangles are congruent in a regular polygon, so multiplying one triangle's area by n gives the total. The tangent function appears because the apothem is the adjacent side of a right triangle formed by the apothem, half a side, and the line to a vertex, with the half-angle 180°/n at the centre. Regular polygon area is a special, efficient case; irregular polygons require the general decomposition strategy from Lesson 5. Connect to the driving question: if a surveyor in Nakuru ever encounters a perfectly regular plot (e.g., a planned settlement or a school garden), this single formula gives the answer immediately.
Lesson 7 **Area of Irregular Polygons: Decomposing the Surveyor's Shamba**	Learners returned to the anchoring Nakuru shamba polygon from Lesson 1, now equipped with all three triangle area formulas. They drew diagonals from a single vertex to decompose the irregular polygon into $(n - 2)$ triangles and identified, for each triangle, which measurements were available: some had a base and perpendicular height, some had two sides and an included angle, and some had only three side lengths. Learners observed that different triangles within the SAME polygon may require DIFFERENT area formulas.	To find the area of an irregular polygon: (1) decompose into $(n - 2)$ triangles via diagonals from one vertex; (2) for each triangle, assess available data and select the appropriate formula — $A = ½bh$ if perpendicular height is known, $A = ½ab \sin C$ if two sides and included angle are known, or Heron's formula if only three sides are known; (3) calculate each triangle area; (4) sum all triangle areas for the total polygon area. The ability to choose the correct formula based on available data is the core skill of this lesson. Pedagogical note: Use a decision-tree poster on the classroom wall that learners reference before each triangle calculation.	The reason different formulas may apply within the same polygon is that the survey data collected in the field depends on the instrument used and the site conditions — sometimes a theodolite provides an angle, sometimes only a measuring tape provides distances. All three formulas are mathematically equivalent (they give the same area for the same triangle), but each requires different inputs. Fluency in formula selection is precisely the professional skill a Nakuru land surveyor must have. Connect to the driving question: learners can now handle any combination of measurements for any sub-triangle within the irregular shamba.
Lesson 8 **Consolidation, Model Revision & Real-World Applications: Answering the Driving Question**	Learners reviewed their cumulative Driving Question Board entries from all seven prior lessons and observed how the toolkit of strategies had grown from a single formula (½bh) to five interconnected strategies. They applied the full toolkit to a final,	A complete polygon area toolkit comprises five strategies: (1) $A = ½bh$ when a perpendicular height is directly available; (2) $A = ½ab \sin C$ when two sides and the included angle are known; (3) Heron's formula $A = \sqrt{s(s-a)(s-b)(s-c)}$ when only	The power of the toolkit lies in its completeness: no matter what combination of side lengths, heights, and angles a Nakuru land surveyor can measure in the field, at least one formula will always apply. This directly and fully answers the driving

	complex Nakuru shamba problem — an irregular hexagonal plot with mixed available data — and compared their computed area to a 'surveyor's reference answer', observing close agreement and discussing sources of rounding error.	three side lengths are known; (4) $A = (n \times s^2) / (4 \times \tan(180°/n))$ for any regular polygon given n and s; (5) polygon decomposition into $(n - 2)$ triangles for any irregular polygon, applying formulas (1)–(3) to each triangle and summing. Strategy selection depends entirely on the measurements available, not on the shape's appearance. Pedagogical note: The final lesson should function as a structured revision; allow learners to use their exercise books and formula sheets so the focus is on strategic thinking, not memorisation.	question. Teachers should facilitate a whole-class 'driving question answer' moment — cold-call three to four learners to articulate the answer in their own words before writing the class consensus answer on the board. Emphasise that mathematics provides exact, reproducible results that protect land ownership rights, linking the subject to citizenship, economic justice, and the real lives of farming communities across Kenya.